Exercises: Multivariate Integration – Answer Key

Mathematical Foundations for Machine Learning

Polar Coordinates and Coordinate Transformations

Exercise 1.1 [Easy]
Consider the polar coordinate transformation $\phi(r, \theta) = (r\cos\theta,\ r\sin\theta)^T$.

(a) Compute the Jacobian matrix $D\phi(r, \theta)$.

(b) Compute $\det(D\phi(r, \theta))$ and interpret geometrically what it represents.

Solution

(a) We differentiate each component with respect to r and θ:

$$D\phi(r,\theta) = \begin{pmatrix} \frac{\partial(r\cos\theta)}{\partial r} & \frac{\partial(r\cos\theta)}{\partial\theta} \\ \frac{\partial(r\sin\theta)}{\partial r} & \frac{\partial(r\sin\theta)}{\partial\theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}.$$

(b) The determinant is:

$$\det(D\phi) = r\cos^2\theta + r\sin^2\theta = r.$$

Geometrically, this means that an infinitesimal area element in polar coordinates is $r\,dr\,d\theta$: regions farther from the origin are stretched more.

Exercise 1.2 [Easy]
Convert the integral $\int\int_D (x^2 + y^2)\,dx\,dy$, where $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le R^2\}$, into polar coordinates. Set up the resulting integral but **do not** evaluate it.

Solution

Under the polar coordinate transformation $x = r\cos\theta$, $y = r\sin\theta$, we have $x^2 + y^2 = r^2$ and $dx\,dy = r\,dr\,d\theta$. The disk $x^2 + y^2 \le R^2$ maps to $0 \le r \le R$, $0 \le \theta < 2\pi$. Thus:

$$\int\int_D (x^2 + y^2)\,dx\,dy = \int_0^{2\pi}\int_0^R r^2 \cdot r\,dr\,d\theta = \int_0^{2\pi}\int_0^R r^3\,dr\,d\theta.$$

Exercise 1.3 [Medium]
Evaluate the integral

$$I = \int\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy$$

using polar coordinates. Then use this to show that

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Finally, derive the normalization constant for the univariate Gaussian density $\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{y^2}{2\sigma^2}}$.

Solution

Step 1: Switch to polar coordinates. With $x^2 + y^2 = r^2$ and $dx\,dy = r\,dr\,d\theta$:

$$I = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r\,dr\,d\theta = 2\pi \int_0^{\infty} re^{-r^2}\,dr.$$

Step 2: Evaluate the radial integral. With substitution $u = r^2$, $du = 2r\,dr$:

$$\int_0^{\infty} re^{-r^2}\,dr = \frac{1}{2}\int_0^{\infty} e^{-u}\,du = \frac{1}{2}.$$

Thus $I = 2\pi \cdot \frac{1}{2} = \pi$.

Step 3: By Fubini's theorem, $I = \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)^2 = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.

Step 4: Substitute $x = \frac{y}{\sqrt{2\sigma^2}}$, so $dx = \frac{1}{\sqrt{2\sigma^2}}\,dy$:

$$\sqrt{\pi} = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \frac{1}{\sqrt{2\sigma^2}}\int_{-\infty}^{\infty} e^{-\frac{y^2}{2\sigma^2}}\,dy$$

which gives $\int_{-\infty}^{\infty} e^{-\frac{y^2}{2\sigma^2}}\,dy = \sqrt{2\pi\sigma^2}$, confirming the normalization.

Exercise 1.4 [Medium]

Consider the transformation $\phi : \mathbb{R}^2_{++} \to \mathbb{R}^2_{++}$ defined by

$$\phi(u, v) = (uv,\ u/v).$$

(a) Compute the Jacobian $D\phi(u, v)$ and its determinant.

(b) Find the inverse mapping $\phi^{-1}(x, y)$. Verify that $\det(D\phi^{-1}(x, y)) = \frac{1}{\det(D\phi(u,v))}$ at corresponding points.

Solution

(a) We have $\phi_1(u, v) = uv$ and $\phi_2(u, v) = u/v$.

$$D\phi(u, v) = \begin{pmatrix} v & u \\ 1/v & -u/v^2 \end{pmatrix}.$$

$$\det(D\phi) = v \cdot (-u/v^2) - u \cdot (1/v) = -u/v - u/v = -2u/v.$$

So $|\det(D\phi)| = 2u/v$.

(b) From $x = uv$ and $y = u/v$, we get $xy = u^2$ and $x/y = v^2$, giving $u = \sqrt{xy}$, $v = \sqrt{x/y}$.

Thus $\phi^{-1}(x, y) = (\sqrt{xy},\ \sqrt{x/y})$.

Now compute $D\phi^{-1}$:

$$D\phi^{-1}(x, y) = \begin{pmatrix} \frac{1}{2}\sqrt{y/x} & \frac{1}{2}\sqrt{x/y} \\ \frac{1}{2\sqrt{xy}} & -\frac{\sqrt{x}}{2y\sqrt{y}} \end{pmatrix}.$$

After simplification, $\det(D\phi^{-1}) = -\frac{1}{2y} = -\frac{v}{2u}$, which equals $\frac{1}{\det(D\phi)} = \frac{1}{-2u/v} = -\frac{v}{2u}$. ✓

Exercise 1.5 [Hard]

Let $\phi : \mathbb{R}^3_+ \to \mathbb{R}^3$ be the spherical coordinate transformation:

$$\phi(r, \theta, \varphi) = (r \sin\theta \cos\varphi,\ r \sin\theta \sin\varphi,\ r \cos\theta)$$

where $r > 0$, $\theta \in [0, \pi]$, $\varphi \in [0, 2\pi)$.

(a) Compute $\det(D\phi(r, \theta, \varphi))$.

(b) Use the transformation formula and spherical coordinates to evaluate

$$\int\int\int_{\mathbb{R}^3} e^{-\|x\|^2}\, dx$$

and hence show that $\int_{-\infty}^{\infty} e^{-t^2}\, dt = \sqrt{\pi}$ by a purely 3D argument.

(c) Using the result of (b), find the normalization constant for the multivariate Gaussian $\mathcal{N}(0, \sigma^2 I_3)$.

Solution

(a) The Jacobian matrix is:

$$D\phi = \begin{pmatrix} \sin\theta\cos\varphi & r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ \sin\theta\sin\varphi & r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix}.$$

Expanding along the third row:

$$\det(D\phi) = \cos\theta \cdot r^2 \sin\theta\cos\theta(\cos^2\varphi + \sin^2\varphi) + r\sin\theta \cdot r\sin\theta(\sin^2\varphi\cos\varphi \cdot \sin\theta + \cdots)$$

After careful expansion: $\det(D\phi) = r^2 \sin\theta$.

(b) By the transformation formula:

$$I_3 = \int_{\mathbb{R}^3} e^{-\|x\|^2}\, dx = \int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} e^{-r^2} r^2 \sin\theta\, dr\, d\theta\, d\varphi.$$

Separating:

$$I_3 = \underbrace{\int_0^{2\pi} d\varphi}_{2\pi} \cdot \underbrace{\int_0^{\pi} \sin\theta\, d\theta}_{2} \cdot \int_0^{\infty} r^2 e^{-r^2}\, dr = 4\pi \cdot \frac{\sqrt{\pi}}{4} = \pi^{3/2},$$

where $\int_0^{\infty} r^2 e^{-r^2}\, dr = \frac{\sqrt{\pi}}{4}$ by integration by parts.

By Fubini, $I_3 = \left(\int_{-\infty}^{\infty} e^{-t^2}\, dt\right)^3 = \pi^{3/2}$, giving $\int_{-\infty}^{\infty} e^{-t^2}\, dt = \pi^{1/2} = \sqrt{\pi}$.

(c) For $\mathcal{N}(0, \sigma^2 I_3)$, we need c such that $c\int_{\mathbb{R}^3} e^{-\frac{\|x\|^2}{2\sigma^2}}\, dx = 1$. Using the substitution $x = \sqrt{2\sigma^2}\, u$, $dx = (2\sigma^2)^{3/2}\, du$:

$$\int_{\mathbb{R}^3} e^{-\frac{\|x\|^2}{2\sigma^2}}\, dx = (2\sigma^2)^{3/2}\pi^{3/2} = (2\pi\sigma^2)^{3/2}.$$

So $c = \frac{1}{(2\pi\sigma^2)^{3/2}}$.

Fubini's Theorem and Iterated Integrals

Exercise 2.1 [Easy]
Let $f(x, y) = x^2 y$ on $K = [0, 2] \times [1, 3]$.

(a) Compute $\int_K f(x, y)\, dx\, dy$ by first integrating over x, then over y.
(b) Compute the same integral by first integrating over y, then over x.
(c) Verify that both orders give the same result. Which theorem guarantees this?

Solution

(a) Integrate x first:

$$\int_1^3 \left(\int_0^2 x^2 y\, dx \right) dy = \int_1^3 y \left[\frac{x^3}{3} \right]_0^2 dy = \int_1^3 \frac{8y}{3}\, dy = \frac{8}{3} \left[\frac{y^2}{2} \right]_1^3 = \frac{8}{3} \cdot 4 = \frac{32}{3}.$$

(b) Integrate y first:

$$\int_0^2 \left(\int_1^3 x^2 y\, dy \right) dx = \int_0^2 x^2 \left[\frac{y^2}{2} \right]_1^3 dx = \int_0^2 4x^2\, dx = 4 \cdot \frac{8}{3} = \frac{32}{3}.$$

(c) Both give $\frac{32}{3}$. This is guaranteed by **Fubini's theorem**: since f is continuous (hence bounded and Riemann integrable) on the compact product set K, the order of integration can be exchanged.

Exercise 2.2 [Medium]
Consider the integral

$$I = \int_0^1 \int_x^1 e^{y^2}\, dy\, dx.$$

(a) Explain why this integral cannot be evaluated directly by first integrating over y.
(b) Sketch the region of integration and swap the order of integration using Fubini's theorem.
(c) Evaluate the resulting integral.

Solution

(a) The antiderivative of e^{y^2} with respect to y cannot be written in terms of elementary functions (it is related to the error function). So direct computation of the inner integral is not feasible.
(b) The region is $\{(x, y) : 0 \le x \le 1,\ x \le y \le 1\} = \{(x, y) : 0 \le y \le 1,\ 0 \le x \le y\}$. Swapping:

$$I = \int_0^1 \int_0^y e^{y^2}\, dx\, dy.$$

(c) The inner integral over x is trivial:

$$I = \int_0^1 y\, e^{y^2}\, dy = \left[\frac{1}{2} e^{y^2} \right]_0^1 = \frac{e - 1}{2}.$$

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x, y, z) = xyz$, and let $K = [0, 1]^3$.

(a) Compute $\int_K f\, d(x,y,z)$ using Fubini's theorem by splitting into three iterated one-dimensional integrals.

(b) Now let $g(x,y,z) = xy + yz$. Compute $\int_K g\, d(x,y,z)$. Explain which property of the integral allows you to split this computation.

Solution

(a) By Fubini (since f is continuous on the compact set K):

$$\int_K xyz\, dx\, dy\, dz = \int_0^1 \left(\int_0^1 \left(\int_0^1 xyz\, dx \right) dy \right) dz.$$

Inner: $\int_0^1 xyz\, dx = \frac{yz}{2}$. Middle: $\int_0^1 \frac{yz}{2}\, dy = \frac{z}{4}$. Outer: $\int_0^1 \frac{z}{4}\, dz = \frac{1}{8}$.
(b) By **linearity** of the integral:

$$\int_K (xy + yz)\, d(x,y,z) = \int_K xy\, d(x,y,z) + \int_K yz\, d(x,y,z).$$

For $\int_K xy\, d(x,y,z)$: the integrand does not depend on z, so $\int_0^1 dz = 1$ and we get $\int_0^1 \int_0^1 xy\, dx\, dy = \frac{1}{4}$.
Similarly $\int_K yz\, d(x,y,z) = 1 \cdot \frac{1}{4} = \frac{1}{4}$. Total: $\frac{1}{2}$.

Exercise 2.4 [Hard]
In Bayesian machine learning, one often needs to compute the **marginal likelihood** by integrating out a latent variable. Suppose a model has a scalar latent variable w and a scalar observation y, with:

$$p(y \mid w) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-w)^2}{2}}, \qquad p(w) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{w^2}{2\sigma^2}}.$$

Compute the marginal likelihood $p(y) = \int_{-\infty}^{\infty} p(y \mid w)\, p(w)\, dw$ by:
(a) Writing out the full integrand and combining exponents.
(b) Completing the square in w.
(c) Using Fubini/variable substitution to evaluate the resulting Gaussian integral.
State which result from the Gaussian integral computation you are using.

Solution

(a) The integrand is:

$$p(y|w)p(w) = \frac{1}{2\pi\sigma} \exp\left(-\frac{(y-w)^2}{2} - \frac{w^2}{2\sigma^2}\right).$$

Combine exponents: $-\frac{(y-w)^2}{2} - \frac{w^2}{2\sigma^2} = -\frac{\sigma^2(y-w)^2+w^2}{2\sigma^2}$.
Expand: $\sigma^2(y-w)^2 + w^2 = (\sigma^2+1)w^2 - 2\sigma^2 yw + \sigma^2 y^2$.
(b) Complete the square in w:

$$(\sigma^2+1)\left(w - \frac{\sigma^2 y}{\sigma^2+1}\right)^2 + \frac{\sigma^2 y^2}{\sigma^2+1}.$$

So the exponent becomes $-\frac{(\sigma^2+1)}{2\sigma^2}\left(w - \frac{\sigma^2 y}{\sigma^2+1}\right)^2 - \frac{y^2}{2(\sigma^2+1)}$.
(c) Factor out the y-dependent term:

$$p(y) = \frac{1}{2\pi\sigma} e^{-\frac{y^2}{2(\sigma^2+1)}} \int_{-\infty}^{\infty} \exp\left(-\frac{(\sigma^2+1)}{2\sigma^2}(w-\mu_w)^2\right) dw.$$

This is a Gaussian integral with variance $\tau^2 = \frac{\sigma^2}{\sigma^2+1}$, so it equals $\sqrt{2\pi\tau^2} = \sigma\sqrt{\frac{2\pi}{\sigma^2+1}}$.
Thus: $p(y) = \frac{1}{\sqrt{2\pi(\sigma^2+1)}} e^{-\frac{y^2}{2(\sigma^2+1)}}$.
This is $\mathcal{N}(0, \sigma^2+1)$, using the normalization $\int e^{-t^2}\,dt = \sqrt{\pi}$.

The Transformation Formula

Exercise 3.1 [Easy]
Let $\phi(x) = Ax + b$ where $A \in \mathbb{R}^{n\times n}$ is invertible and $b \in \mathbb{R}^n$. Let $K \subset \mathbb{R}^n$ be compact.
Using the transformation formula, show that

$$\mathrm{vol}(\phi(K)) = |\det(A)| \cdot \mathrm{vol}(K).$$

Solution

By the transformation formula (Theorem 65), for $f \equiv 1$:

$$\mathrm{vol}(\phi(K)) = \int_{\phi(K)} 1\,dy = \int_K 1 \cdot |\det(D\phi(x))|\,dx.$$

Since $\phi(x) = Ax + b$, we have $D\phi(x) = A$ for all x, so $|\det(D\phi(x))| = |\det(A)|$ is constant.
Thus:

$$\mathrm{vol}(\phi(K)) = |\det(A)| \int_K dx = |\det(A)| \cdot \mathrm{vol}(K). \qquad \square$$

Exercise 3.2 [Easy]
Evaluate $I = \int\int_D e^{-(x^2+y^2)}\,dx\,dy$ where $D = \{(x,y) : 1 \le x^2+y^2 \le 4\}$ is the annular region

between circles of radius 1 and 2, using the transformation formula with polar coordinates.

Solution

Apply the transformation formula with $\phi(r,\theta) = (r\cos\theta, r\sin\theta)$ and $|\det(D\phi)| = r$. The preimage is $K = [1,2] \times [0,2\pi)$:

$$I = \int_K e^{-r^2} \cdot r\, dr\, d\theta = \int_0^{2\pi} d\theta \int_1^2 re^{-r^2}\, dr = 2\pi \left[-\frac{1}{2}e^{-r^2}\right]_1^2 = 2\pi \cdot \frac{1}{2}(e^{-1} - e^{-4}) = \pi(e^{-1} - e^{-4}).$$

Exercise 3.3 [Medium]
Consider the linear transformation $\phi(u,v) = (2u+v,\ u+3v)$.

(a) Compute $|\det(D\phi)|$.

(b) Let $S = \{(u,v) : 0 \le u \le 1,\ 0 \le v \le 1\}$ be the unit square. Use the transformation formula to compute

$$\int_{\phi(S)} (x-y)\, dx\, dy.$$

Solution

(a) $D\phi = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, so $|\det(D\phi)| = |6-1| = 5$.

(b) By the transformation formula:

$$\int_{\phi(S)} (x-y)\, dx\, dy = \int_S (\phi_1(u,v) - \phi_2(u,v)) \cdot |\det(D\phi)|\, du\, dv.$$

We have $\phi_1 - \phi_2 = (2u+v) - (u+3v) = u - 2v$, so:

$$= 5\int_0^1 \int_0^1 (u-2v)\, du\, dv = 5\int_0^1 \left[\frac{u^2}{2} - 2vu\right]_0^1 dv = 5\int_0^1 \left(\frac{1}{2} - 2v\right) dv$$

$$= 5\left[\frac{v}{2} - v^2\right]_0^1 = 5\left(\frac{1}{2} - 1\right) = -\frac{5}{2}.$$

Exercise 3.4 [Medium]
Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be a C^1-diffeomorphism with Jacobian $D\phi(x)$ at point x.

(a) State the relationship between $D(\phi^{-1})(y)$ and $D\phi(x)$ at $y = \phi(x)$.

(b) Prove this relationship using the chain rule applied to $\phi^{-1}(\phi(x)) = x$.

(c) Show that $\det(D(\phi^{-1})(y)) = \frac{1}{\det(D\phi(x))}$ at corresponding points.

Solution

(a) The relationship is $D(\phi^{-1})(\phi(x)) = (D\phi(x))^{-1}$.
(b) Starting from the identity $\phi^{-1}(\phi(x)) = x$, differentiate both sides with respect to x using the chain rule:

$$D(\phi^{-1})(\phi(x)) \cdot D\phi(x) = I.$$

Since $D\phi(x)$ is invertible (as ϕ is a diffeomorphism), multiply both sides on the right by $(D\phi(x))^{-1}$:

$$D(\phi^{-1})(\phi(x)) = (D\phi(x))^{-1}.$$

(c) Taking the determinant of both sides of the relation in (a):

$$\det(D(\phi^{-1})(y)) = \det((D\phi(x))^{-1}) = \frac{1}{\det(D\phi(x))}.$$

This uses the property $\det(M^{-1}) = 1/\det(M)$ for any invertible matrix M. $\square$

Exercise 3.5 [Hard]
The **volume of the n-dimensional unit ball** $B_n = \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$ can be computed as $V_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$.

Using the transformation formula with the substitution $x = ru$ where $u \in \partial B_n$ (the unit sphere) and $r \geq 0$:

(a) Explain why $\mathrm{vol}(B_n(0, R)) = R^n \cdot V_n$ using a simple scaling argument via the transformation formula.

(b) Use part (a) to prove the volume concentration result:

$$\frac{\mathrm{vol}(B_n(0, 1-\varepsilon))}{\mathrm{vol}(B_n(0, 1))} = (1-\varepsilon)^n \xrightarrow{n\to\infty} 0$$

for any fixed $0 < \varepsilon < 1$. Give an explicit bound on n such that this ratio is less than δ for given $\delta > 0$.

(c) Discuss the implication for high-dimensional machine learning: why does this mean that data sampled from a uniform distribution on B_n is concentrated near the boundary?

Solution

(a) Apply the transformation formula with the scaling map $\phi(x) = Rx$, so $D\phi = R \cdot I_n$ and $|\det(D\phi)| = R^n$:

$$\mathrm{vol}(B_n(0,R)) = \int_{B_n(0,R)} 1\,dy = \int_{B_n(0,1)} 1 \cdot R^n\,dx = R^n \cdot V_n.$$

(b) Therefore:

$$\frac{\mathrm{vol}(B_n(0,1-\varepsilon))}{\mathrm{vol}(B_n(0,1))} = \frac{(1-\varepsilon)^n V_n}{V_n} = (1-\varepsilon)^n.$$

Since $0 < 1-\varepsilon < 1$, we have $(1-\varepsilon)^n \to 0$ as $n \to \infty$.
For the explicit bound: we need $(1-\varepsilon)^n < \delta$. Taking logarithms: $n\ln(1-\varepsilon) < \ln\delta$, so

$$n > \frac{\ln\delta}{\ln(1-\varepsilon)} = \frac{\ln(1/\delta)}{\ln(1/(1-\varepsilon))}.$$

(c) This means that the **thin outer shell** $B_n(0,1) \setminus B_n(0,1-\varepsilon)$ contains almost all the volume of the ball in high dimensions. For ML, this implies that uniformly sampled data points in a high-dimensional ball will almost all lie near the surface. This is a fundamental manifestation of the **curse of dimensionality**: distances between points become less informative, and algorithms that rely on volume-based reasoning (like nearest-neighbor) need exponentially more data as n grows.

Jacobian of the Inverse

Exercise 4.1 [Easy]
Let $\phi : \mathbb{R} \to \mathbb{R}$ be defined by $\phi(x) = e^x$.

(a) Compute $\phi'(x)$.
(b) Find $\phi^{-1}(y)$ and compute $(\phi^{-1})'(y)$ directly.
(c) Verify that $(\phi^{-1})'(y) = \frac{1}{\phi'(x)}$ at $y = \phi(x)$.

Solution

(a) $\phi'(x) = e^x$.
(b) $\phi^{-1}(y) = \ln(y)$, so $(\phi^{-1})'(y) = \frac{1}{y}$.
(c) At $y = \phi(x) = e^x$:

$$\frac{1}{\phi'(x)} = \frac{1}{e^x} = \frac{1}{y}.$$

This matches $(\phi^{-1})'(y) = \frac{1}{y}$. ✓
This is the one-dimensional version of the general result $D(\phi^{-1})(\phi(x)) = (D\phi(x))^{-1}$.

Exercise 4.2 [Medium]
Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $\phi(u,v) = (e^u \cos v,\ e^u \sin v)$.

(a) Compute $D\phi(u,v)$ and show that it is invertible for all (u,v).
(b) Without explicitly finding ϕ^{-1}, compute $D(\phi^{-1})(x,y)$ at $(x,y) = \phi(u,v)$ using the Jacobian-of-the-inverse theorem.

(c) What does the inverse function theorem (Theorem 63) guarantee about ϕ near any point (u_0, v_0)?

Solution

(a)

$$D\phi(u,v) = \begin{pmatrix} e^u \cos v & -e^u \sin v \\ e^u \sin v & e^u \cos v \end{pmatrix}.$$

Determinant: $\det(D\phi) = e^{2u}(\cos^2 v + \sin^2 v) = e^{2u} > 0$ for all (u,v). So $D\phi$ is always invertible.

(b) By the Jacobian-of-the-inverse theorem:

$$D(\phi^{-1})(\phi(u,v)) = (D\phi(u,v))^{-1} = \frac{1}{e^{2u}} \begin{pmatrix} e^u \cos v & e^u \sin v \\ -e^u \sin v & e^u \cos v \end{pmatrix} = \frac{1}{e^u} \begin{pmatrix} \cos v & \sin v \\ -\sin v & \cos v \end{pmatrix}.$$

Expressing in terms of (x,y): since $x^2 + y^2 = e^{2u}$, we get $e^u = \sqrt{x^2+y^2}$, so:

$$D(\phi^{-1})(x,y) = \frac{1}{x^2+y^2} \begin{pmatrix} x & y \\ -y & x \end{pmatrix}.$$

(c) The inverse function theorem guarantees that near any (u_0, v_0), ϕ restricted to a neighborhood U of (u_0, v_0) is a C^r-diffeomorphism onto a neighborhood V of $\phi(u_0, v_0)$. Since $\det(D\phi) \neq 0$ everywhere, this holds at every point. (Note: the theorem provides a *local* inverse; global invertibility requires additional arguments.)

Exercise 4.3 [Hard]

In normalizing flows (a generative ML model), one transforms a simple base distribution $p_Z(z)$ (e.g., Gaussian) through an invertible map $f : \mathbb{R}^d \to \mathbb{R}^d$ to obtain a complex distribution $p_X(x)$ for $x = f(z)$.

The **change of variables formula** for probability densities gives:

$$p_X(x) = p_Z(f^{-1}(x)) \cdot |\det(D(f^{-1})(x))|.$$

(a) Using the Jacobian-of-the-inverse theorem, rewrite this as:

$$p_X(x) = p_Z(z) \cdot |\det(Df(z))|^{-1}$$

where $z = f^{-1}(x)$. Justify each step.

(b) Suppose f is a composition of K invertible maps: $f = f_K \circ f_{K-1} \circ \cdots \circ f_1$. Show that

$$\log p_X(x) = \log p_Z(z_0) - \sum_{k=1}^{K} \log |\det(Df_k(z_{k-1}))|$$

where $z_0 = f^{-1}(x)$ and $z_k = f_k(z_{k-1})$.

(c) Explain why computing $\det(Df_k)$ for each layer is the computational bottleneck and why architectures like RealNVP use transformations with **triangular** Jacobians.

Solution

(a) Start with the change of variables formula:

$$p_X(x) = p_Z(f^{-1}(x)) \cdot |\det(D(f^{-1})(x))|.$$

By the Jacobian-of-the-inverse theorem: $D(f^{-1})(x) = (Df(z))^{-1}$ where $z = f^{-1}(x)$.
Taking determinants: $\det(D(f^{-1})(x)) = \det((Df(z))^{-1}) = \frac{1}{\det(Df(z))}$.
Substituting and noting $f^{-1}(x) = z$:

$$p_X(x) = p_Z(z) \cdot |\det(Df(z))|^{-1}. \qquad \square$$

(b) Taking logarithms of the result in (a): $\log p_X(x) = \log p_Z(z) - \log|\det(Df(z))|$.
For $f = f_K \circ \cdots \circ f_1$, the chain rule gives $Df(z_0) = Df_K(z_{K-1}) \cdots Df_1(z_0)$. Taking determinants:

$$\det(Df(z_0)) = \prod_{k=1}^{K} \det(Df_k(z_{k-1})),$$

so $\log|\det(Df(z_0))| = \sum_{k=1}^{K} \log|\det(Df_k(z_{k-1}))|$.
Substituting with $z_0 = f^{-1}(x)$:

$$\log p_X(x) = \log p_Z(z_0) - \sum_{k=1}^{K} \log|\det(Df_k(z_{k-1}))|. \qquad \square$$

(c) Computing the determinant of a general $d \times d$ matrix costs $O(d^3)$. In deep models with d possibly being thousands (image pixels), this is prohibitive.
Triangular Jacobians (as in RealNVP) have the property that $\det(J) = \prod_i J_{ii}$, reducing the cost to $O(d)$. These architectures achieve triangularity by designing coupling layers where each output dimension depends on inputs in a structured way, making the Jacobian lower- or upper-triangular by construction.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix with eigendecomposition $A = Q\Lambda Q^T$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_i > 0$.

Consider the integral

$$I = \int_{\mathbb{R}^n} e^{-\frac{1}{2}x^T Ax}\, dx.$$

(a) Apply the transformation formula with $x = Qy$ (where Q is orthogonal). What is $|\det(DQ)|$? Show that the integral becomes $\int_{\mathbb{R}^n} e^{-\frac{1}{2}y^T \Lambda y}\, dy$.

(b) Use Fubini's theorem to split this into n independent one-dimensional Gaussian integrals and show that

$$I = \frac{(2\pi)^{n/2}}{\sqrt{\det(A)}}.$$

(c) Use this result to verify that the multivariate Gaussian $\mathcal{N}(\mu, \Sigma)$ density

$$p(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

integrates to 1.

Solution

(a) The map $\phi(y) = Qy$ is linear with $D\phi = Q$ and $|\det(Q)| = 1$ (since Q is orthogonal: $\det(Q)^2 = \det(Q^TQ) = \det(I) = 1$).
By the transformation formula:

$$I = \int_{\mathbb{R}^n} e^{-\frac{1}{2}x^TAx}\,dx = \int_{\mathbb{R}^n} e^{-\frac{1}{2}(Qy)^TA(Qy)} \cdot 1\,dy.$$

Since $A = Q\Lambda Q^T$: $(Qy)^TA(Qy) = y^TQ^TQ\Lambda Q^TQy = y^T\Lambda y$.
Thus $I = \int_{\mathbb{R}^n} e^{-\frac{1}{2}y^T\Lambda y}\,dy$.
(b) Since Λ is diagonal: $y^T\Lambda y = \sum_{i=1}^n \lambda_i y_i^2$. By Fubini:

$$I = \prod_{i=1}^n \int_{-\infty}^{\infty} e^{-\frac{\lambda_i}{2}y_i^2}\,dy_i = \prod_{i=1}^n \sqrt{\frac{2\pi}{\lambda_i}} = (2\pi)^{n/2}\prod_{i=1}^n \lambda_i^{-1/2} = \frac{(2\pi)^{n/2}}{\sqrt{\det(A)}},$$

using $\det(A) = \prod_i \lambda_i$.
(c) Substitute $u = x - \mu$ (a translation, $|\det| = 1$) and set $A = \Sigma^{-1}$:

$$\int_{\mathbb{R}^n} p(x)\,dx = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}}\int_{\mathbb{R}^n} e^{-\frac{1}{2}u^T\Sigma^{-1}u}\,du = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \cdot \frac{(2\pi)^{n/2}}{\sqrt{\det(\Sigma^{-1})}}.$$

Since $\det(\Sigma^{-1}) = 1/\det(\Sigma) = 1/|\Sigma|$:

$$= \frac{(2\pi)^{n/2}|\Sigma|^{1/2}}{(2\pi)^{n/2}|\Sigma|^{1/2}} = 1. \qquad \square$$

Exercise 5.2 [Hard]
(Change of variables for probability densities) Let $Z \sim \mathcal{N}(0, I_2)$ and define the transformation $\phi(z_1, z_2) = (z_1, z_1 + z_2)$.

(a) Compute $D\phi$ and $|\det(D\phi)|$.

(b) Find $\phi^{-1}(x_1, x_2)$ and compute $D(\phi^{-1})$ both directly and via the Jacobian-of-the-inverse theorem.

(c) Using the change of variables formula for densities, find $p_X(x)$ where $X = \phi(Z)$.

(d) Identify the distribution of $X = (X_1, X_2)^T$. What is its covariance matrix? Verify by computing $\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(Z_1, Z_1 + Z_2)$ directly.

Solution

(a) $D\phi = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $|\det(D\phi)| = 1$.

(b) From $x_1 = z_1$, $x_2 = z_1 + z_2$: $\phi^{-1}(x_1, x_2) = (x_1, x_2 - x_1)$.

Directly: $D(\phi^{-1}) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$.

Via the theorem: $(D\phi)^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$. ✓

(c) The density of X:

$$p_X(x) = p_Z(\phi^{-1}(x)) \cdot |\det(D(\phi^{-1})(x))| = \frac{1}{2\pi} e^{-\frac{1}{2}(x_1^2 + (x_2 - x_1)^2)} \cdot 1.$$

Expanding: $x_1^2 + (x_2 - x_1)^2 = 2x_1^2 - 2x_1x_2 + x_2^2$.
So $p_X(x) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}(2x_1^2 - 2x_1x_2 + x_2^2)\right)$.

(d) This is $\mathcal{N}(0, \Sigma)$ where we need $\Sigma^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$, giving $\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.

Direct verification: $\mathrm{Var}(X_1) = \mathrm{Var}(Z_1) = 1$, $\mathrm{Var}(X_2) = \mathrm{Var}(Z_1 + Z_2) = 2$, $\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(Z_1, Z_1 + Z_2) = \mathrm{Var}(Z_1) + \mathrm{Cov}(Z_1, Z_2) = 1 + 0 = 1$. This matches Σ. ✓

Exercise 5.3 [Hard]
(Expectation via the transformation formula) Let $X \sim \mathcal{N}(0, \sigma^2)$ be a univariate Gaussian. Using polar coordinates in $\mathbb{R}^2$ and Fubini's theorem:

(a) Show that $\mathbb{E}[X^2] = \sigma^2$ by computing

$$\mathbb{E}[X^2] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2\sigma^2}}\, dx$$

via the trick of writing $\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]$ for i.i.d. copies X, Y, converting to polar coordinates, and factoring.

(b) More generally, show that for $X \sim \mathcal{N}(0, \sigma^2 I_n)$, the transformation formula and Fubini give $\mathbb{E}[\|X\|^2] = n\sigma^2$.

Solution

(a) Let X, Y be i.i.d. $\mathcal{N}(0, \sigma^2)$. Consider:

$$\mathbb{E}[X^2] \cdot 1 = \mathbb{E}[X^2] \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \int e^{-y^2/(2\sigma^2)} dy = \frac{1}{2\pi\sigma^2} \int \int x^2 e^{-(x^2+y^2)/(2\sigma^2)}\, dx\, dy.$$

Switch to polar coordinates ($x = r\cos\theta$, $dx\, dy = r\, dr\, d\theta$):

$$= \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \int_0^{\infty} r^2 \cos^2\theta \cdot e^{-r^2/(2\sigma^2)} r\, dr\, d\theta.$$

Separate by Fubini: $\int_0^{2\pi} \cos^2\theta\, d\theta = \pi$. For the radial part, substituting $u = r^2/(2\sigma^2)$:

$$\int_0^{\infty} r^3 e^{-r^2/(2\sigma^2)}\, dr = 2\sigma^4.$$

Thus $\mathbb{E}[X^2] = \frac{1}{2\pi\sigma^2} \cdot \pi \cdot 2\sigma^4 = \sigma^2$.
(b) For $X = (X_1, \ldots, X_n) \sim \mathcal{N}(0, \sigma^2 I_n)$, Fubini gives independence of coordinates:

$$\mathbb{E}[\|X\|^2] = \mathbb{E}\left[\sum_{i=1}^n X_i^2\right] = \sum_{i=1}^n \mathbb{E}[X_i^2] = n\sigma^2,$$

where we used linearity of expectation and the fact that each marginal $X_i \sim \mathcal{N}(0, \sigma^2)$ contributes σ^2 by part (a). The key step uses Fubini to factor the joint density into a product of marginals: $p(x) = \prod_{i=1}^n p(x_i)$, which is valid because $\Sigma = \sigma^2 I$ is diagonal.